

NU UZ – FRONT THERAPEUTIC SHOE

CAD DESIGN SHEET (FOR MODELING)

Project: Nu Uz Therapeutic Front Shoe

Applies To: Left Front & Right Front (Fronts only)

Units: millimeters (mm)

1. Planform Size (LOCKED)

- **Overall Shoe Width:** 122.3 mm
- **Overall Shoe Length:** 128.1 mm

2. Breakover Geometry (MANDATORY)

- **Breakover Angle:** 30°
- **Breakover Setback:** 8.0 mm from toe
- **Effective Breakover Width:** ~70 mm

Design intent:

Open / rolled toe with material removed to create early breakover.

Do **not** implement as a simple chamfer.

3. Heel Wedge Geometry (INITIAL CONFIG)

- **Toe Thickness:** ~10 mm
- **Heel Thickness:** ~16 mm
- **Net Wedge:** +6 mm toe → heel
- Wedge is **continuous and smooth** across shoe length
- No abrupt steps
- Bottom surface reflects wedge (not top-only)

4. Sole / Internal Geometry

- Defined **sole relief** throughout
- No unintended solar loading
- Internal frog / caudal features must:
 - Be **broad and forgiving**
 - Allow trimming/removal by farrier
 - Maintain **3–4 mm clearance** from live frog
 - Avoid focal apex pressure

5. Load Path Requirements

- Primary load via **perimeter / wall seating**
- No central sole load
- No frog apex load unless explicitly modified by farrier

6. Adhesive Compatibility

- Glue-on only (MMA)

- Bonding surfaces compatible with MMA cure
- Avoid enclosed voids or sharp adhesive traps

7. Medial / Lateral Adhesive Tabs (IMPORTANT)

Quantity

- **Two (2)** tabs total (approximately centered at 10 and 2 oclock)
- Symmetric: **medial + lateral quarters**

Tab Geometry

- Width (circumferential): **~30 mm**
- Height (vertical): **~20 mm**
- **Solid** (no cutouts)
- Thickness follows local shoe thickness
- 70° angle relative to top of horseshoe

Placement

- Tabs are on the **TOP SURFACE** of the shoe
- Positioned at **medial and lateral quarters**
- Tabs **do NOT extend beyond the outer perimeter**
- Tabs are **NOT toe clips**
- Tabs are **NOT load-bearing**

Function

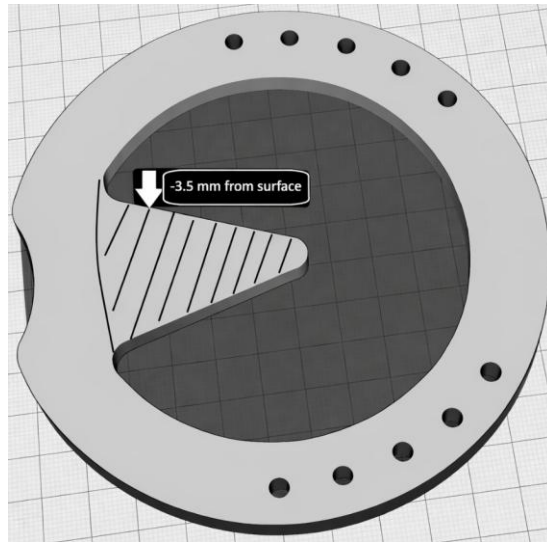
- Adhesive retention & positional stability only

9. Deliverables Requested

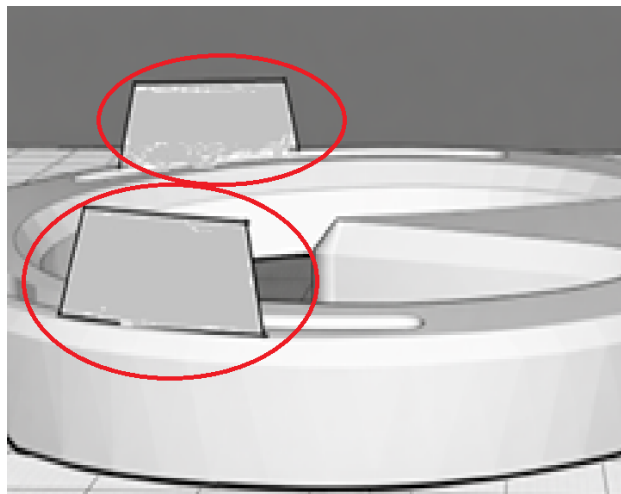
- One solid CAD model (parametric preferred)
- Screenshots:
 - Top view
 - Side view (showing wedge)
 - Isometric view (showing tabs)

Images:

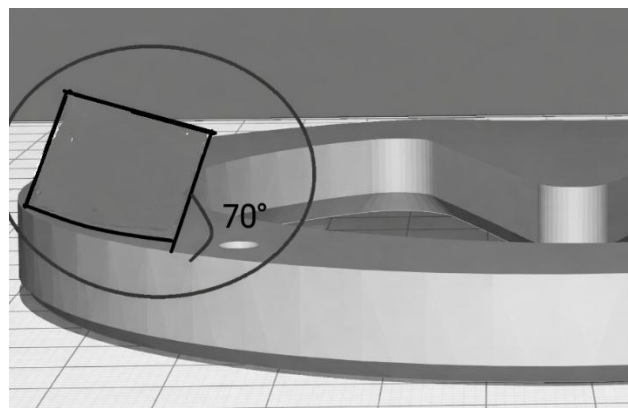
Frog clearance from horseshoe



Tabs on Medial Lateral sides of horseshoes (centers on 2 and 10 o'clock)



Angle of Tab to Shoe



Breakover angle on toe of shoe – 30°

